Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Further Methods in Calculus

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Differentiate $\arcsin x$. **[2]**

2. Differentiate $\arccos x$. **[2]**

3. Differentiate $\arctan x$. **[2]**

4. Find $\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$. **[2]**

5. Find $\int \frac{1}{1+x^2}\,\mathrm{d}x$. **[2]**

6. Differentiate $\operatorname{arsinh} x$. **[2]**

7. Use the substitution $x = \sin\theta$ to find $\int \sqrt{1-x^2}\,\mathrm{d}x$. **[5]**

8. Use the substitution $x = \tan\theta$ to find $\int \frac{1}{1+x^2}\,\mathrm{d}x$. **[3]**

9. Find $\dfrac{d}{dx}\big(\ln|\sec x + \tan x|\big)$. **[3]**

10. Find $\int \sec x\,\mathrm{d}x$. **[3]**

11. Differentiate $x\arcsin x$. **[3]**

12. Find $\int x \arctan x \, \mathrm{d}x$. **[4]**

13. Use partial fractions to integrate $\int \frac{1}{x^2-1} \, \mathrm{d}x$. **[3]**

14. Find $\int \frac{2x+3}{x^2+1} \, \mathrm{d}x$. **[3]**

15. State the formula for integration by parts. **[2]**

Section B – Hard (Q16–25)

16. Find $\int \frac{1}{(1+x^2)^{3/2}} \, \mathrm{d}x$. **[5]**

17. Use the substitution $t = \tan \frac{x}{2}$ to find $\int \frac{1}{\sin x} \, \mathrm{d}x$. **[6]**

18. Find a reduction formula for $I_n = \int_0^{\pi/2} \sin^n x \, \mathrm{d}x$. **[5]**

19. Evaluate $I_5 = \int_0^{\pi/2} \sin^5 x \, \mathrm{d}x$ using a reduction formula. **[4]**

20. Find $\int \sqrt{x^2 + a^2} \, \mathrm{d}x$ using a hyperbolic or trig substitution. **[6]**

21. Differentiate $\operatorname{arsech} x$ and simplify. **[4]**

22. Find $\int \frac{x^2}{1+x^2} \, \mathrm{d}x$. **[3]**

23. Use integration by parts twice to find $\int e^{ax} \sin bx \, \mathrm{d}x$. **[5]**

24. Find the mean value of $\sin^2 x$ over $[0, \pi]$. **[4]**

25. Show that $\int_0^1 \frac{\ln x}{1+x} \, \mathrm{d}x = -\frac{\pi^2}{12}$ (or set up series method). **[5]**

End of Worksheet • *Show all working clearly.*